Worksheet 4.4 — Probability Basics and Venn Diagrams

Mathematics: Analysis & Approaches SL (IB) · Unit 4: Statistics and Probability

The Big Ideas

Venn diagrams and the addition rule solve "at least one" and unknown-overlap problems.

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- Independent: $P(A \cap B) = P(A)P(B)$. Mutually exclusive: $P(A \cap B) = 0$.

Worked Examples

Example 1 — Addition rule

Of 40 students, 24 study French, 18 study Spanish, 9 study both. Find $P(\text{at least one})$.

Solution. $P(F \cup S) = \frac{24}{40} + \frac{18}{40} - \frac{9}{40} = \frac{33}{40}$.

Conclusion: $\frac{33}{40}$.

Example 2 — Unknown overlap

Of 60 people, 38 like tea, 30 like coffee, 8 like neither. Find how many like both.

Solution. At least one $= 52$. $52 = 38 + 30 - \text{both} \Rightarrow \text{both} = 16$.

Conclusion: 16.

Example 3 — Independence check

$P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Are A, B independent?

Solution. $P(A)P(B) = 0.2$, matches $P(A \cap B)$.

Conclusion: yes, independent.

Example 4 — Mutually exclusive check

$P(C) = 0.35$, $P(D) = 0.4$, $P(C \cup D) = 0.75$. Are C, D mutually exclusive?

Solution. $P(C \cap D) = 0.35 + 0.4 - 0.75 = 0$.

Conclusion: yes, $P(C \cap D) = 0$.

Example 5 — Finding a region only in one set

Of 70 people, 45 like hiking, 30 like biking, 12 like both. Find $P(\text{hiking only})$.

Solution. Hiking only $= 45 - 12 = 33$.

Conclusion: $P(\text{hiking only}) = \frac{33}{70}$.

Example 6 — Neither region from a Venn diagram

Of 90 people, 50 like jazz, 40 like classical, 20 like both. Find $P(\text{neither})$.

Solution. At least one $= 50 + 40 - 20 = 70$; neither $= 90 - 70 = 20$.

Conclusion: $P(\text{neither}) = \frac{20}{90} = \frac{2}{9}$.

Example 7 — Three-fact independence verification

A die is rolled and a coin is flipped. Let A="die shows 4" and B="coin shows heads". Verify A, B are independent.

Solution. $P(A) = \frac{1}{6}$, $P(B) = \frac{1}{2}$, $P(A \cap B) = \frac{1}{12}$ (one specific outcome out of 12 equally likely).

Conclusion: $P(A)P(B) = \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12} = P(A \cap B)$ — confirmed independent (as expected, since a die and coin don't influence each other).

Example 8 — Solving for a probability given independence

A, B are independent, $P(A) = 0.6$, $P(A \cup B) = 0.88$. Find $P(B)$.

Solution. $P(A \cup B) = P(A) + P(B) - P(A)P(B) \Rightarrow 0.88 = 0.6 + P(B) - 0.6P(B)$.

$0.88 = 0.6 + 0.4P(B) \Rightarrow 0.28 = 0.4P(B)$.

Conclusion: $P(B) = 0.7$.

Example 9 — Application — overlapping subscriptions

Of 500 households, 320 subscribe to streaming service X, 210 to service Y, and 80 subscribe to neither. Find how many subscribe to both, and P(exactly one).

Solution. At least one $= 500 - 80 = 420$; both $= 320 + 210 - 420 = 110$.

Conclusion: both $= 110$; exactly one $= 420 - 110 = 310 \Rightarrow P = \frac{310}{500} = 0.62$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

$P(A) = 0.6$, $P(B) = 0.25$, $P(A \cap B) = 0.1$. Find $P(A \cup B)$.

Question 2

Of 50 people, 30 own a car, 22 own a bike, 10 own neither. Find how many own both.

Question 3

$P(A) = 0.4$, $P(B) = 0.3$, $P(A \cap B) = 0.12$. Are A, B independent?

Question 4

$P(C) = 0.5$, $P(D) = 0.3$. If mutually exclusive, find $P(C \cup D)$.

Question 5

Of 80 gym members, 50 use weights, 35 use cardio, everyone uses at least one. Find how many use both.

Question 6 — Challenge

A survey of 100 finds 65 like pop music, 45 like rock, and x like both, with 15 liking neither. Find x and $P(\text{pop only})$.

Question 7

Explain why two mutually exclusive events with nonzero probability can never be independent.

Question 8

Of 60 people, 40 like tea, 25 like coffee, 10 like both. Find $P(\text{tea only})$.

Question 9

Of 100 people, 55 like reading, 45 like gaming, 20 like both. Find $P(\text{neither})$.

Question 10

A card is drawn from a deck and a coin flipped. Let A="card is a King", B="coin shows tails". Verify A, B are independent.

Question 11 — Challenge

A, B are independent, $P(A) = 0.5$, $P(A \cup B) = 0.75$. Find $P(B)$.

Question 12 — Challenge

Of 600 households, 380 subscribe to service X, 250 to service Y, 90 subscribe to neither. Find how many subscribe to both, and P(exactly one).

Question 13 — Challenge

$P(A) = 0.35$, $P(B) = 0.5$, $P(A \cup B) = 0.675$. Determine whether A, B are independent.

Question Recap & Answer Key

Question Recap

1. $P(A) = 0.6$, $P(B) = 0.25$, $P(A \cap B) = 0.1$. Find $P(A \cup B)$.
2. Of 50 people, 30 own a car, 22 own a bike, 10 own neither. Find how many own both.
3. $P(A) = 0.4$, $P(B) = 0.3$, $P(A \cap B) = 0.12$. Are A, B independent?
4. $P(C) = 0.5$, $P(D) = 0.3$. If mutually exclusive, find $P(C \cup D)$.
5. Of 80 gym members, 50 use weights, 35 use cardio, everyone uses at least one. Find how many use both.
6. A survey of 100 finds 65 like pop music, 45 like rock, and x like both, with 15 liking neither. Find x and $P(\text{pop only})$.
7. Explain why two mutually exclusive events with nonzero probability can never be independent.
8. Of 60 people, 40 like tea, 25 like coffee, 10 like both. Find $P(\text{tea only})$.
9. Of 100 people, 55 like reading, 45 like gaming, 20 like both. Find $P(\text{neither})$.
10. A card is drawn from a deck and a coin flipped. Let A="card is a King", B="coin shows tails". Verify A, B are independent.
11. A, B are independent, $P(A) = 0.5$, $P(A \cup B) = 0.75$. Find $P(B)$.
12. Of 600 households, 380 subscribe to service X, 250 to service Y, 90 subscribe to neither. Find how many subscribe to both, and $P(\text{exactly one})$.
13. $P(A) = 0.35$, $P(B) = 0.5$, $P(A \cup B) = 0.675$. Determine whether A, B are independent.

✓ Answer Key

1. $0.6 + 0.25 - 0.1 = 0.75$
2. at least one $= 40$; both $= 30 + 22 - 40 = 12$
3. $P(A)P(B) = 0.12$, matches: yes independent
4. 0.8
5. $50 + 35 - 80 = 5$
6. at least one $= 85$; $85 = 65 + 45 - x \Rightarrow x = 25$; pop only $= 65 - 25 = 40 \Rightarrow P = 0.40$
7. if mutually exclusive, one occurring guarantees the other didn't ($P(A \cap B) = 0$), which contradicts independence unless one probability is already 0
8. tea only $= 40 - 10 = 30 \Rightarrow P = \frac{30}{60} = 0.5$
9. at least one $= 55 + 45 - 20 = 80$; neither $= 100 - 80 = 20 \Rightarrow P = 0.2$
10. $P(A) = \frac{4}{52} = \frac{1}{13}$, $P(B) = \frac{1}{2}$, $P(A \cap B) = \frac{1}{26}$; $P(A)P(B) = \frac{1}{26}$, matches: independent
11. $0.75 = 0.5 + P(B) - 0.5P(B) \Rightarrow 0.25 = 0.5P(B) \Rightarrow P(B) = 0.5$
12. at least one $= 510$; both $= 380 + 250 - 510 = 120$; exactly one $= 510 - 120 = 390 \Rightarrow P = 0.65$
13. $P(A)P(B) = 0.175$; $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.35 + 0.5 - 0.675 = 0.175$: matches, so independent